Requirement Analysis and Feasibility Study

POWER WORLD GYM - KIRIBATHGODA

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Introduction

The proposed Gym Management System is a comprehensive digital solution designed to replace the existing manual and semi-digital processes at Power World Gym, Kiribathgoda. The current system, reliant on physical cards, paper-based logs, and static schedules, is inefficient, prone to error, and limits member engagement and administrative oversight.

This new integrated system aims to streamline all core gym operations into a single, secure, and user-friendly platform. Its primary objective is to automate key processes such as member access, attendance tracking, subscription management, and appointment scheduling. By digitizing these workflows, the system enhances operational accuracy, improves security, enriches the member experience through self-service features and progress tracking, and provides management with valuable, data-driven insights for strategic decision-making. The system is designed to be scalable, reliable, and accessible via various endpoints like mobile devices, and web portals.

Proposed System Functions Overview

The core functions of the proposed system, as derived from the use cases, are categorized as follows:

A. Member Access & Attendance Management:

- QR-Based Access Control (UC-01): Members scan a secure QR code at the entrance for identity verification, automated subscription validation, and door unlocking.
- Automated Subscription Validation (UC-02): A backend service that checks membership status and entitlements in real-time upon access attempts or booking requests.
- Authentication Service (UC-03): Verifies user identity during the check-in/out process.
- Timestamp Logging (UC-04): Precisely records all member arrivals and departures, preventing duplicates and ensuring accurate attendance data.
- Staff Access (UC-09): Manages secure entry and exit for staff members, integrating with the same logging system.

B. Membership & Subscription Management:

- Subscription Purchase & Renewal (UC-05): Allows members to seamlessly buy or renew subscriptions online via integrated payment gateways, with automatic activation and confirmation.

C. Scheduling & Appointment Management:

- Trainer Appointments (UC-06): Enables members to view instructor availability, book sessions, and receive automated confirmations and reminders, with checks for subscription eligibility.
- Workout Schedule Modification (UC-10): Allows members and instructors to edit workout plans, with system checks for conflicts and trainer availability.

D. Member Self-Service & Engagement:

- Workout Completion Tracking (UC-07): Allows members to mark workouts as complete, triggering progress updates.
- Progress Monitoring & Analytics (UC-08): Provides members with a dashboard displaying their workout history, trends, consistency metrics, and personal achievements.

E. Administrative & Management Functions:

- Insights Dashboard (UC-11): Provides instructors and managers with high-level analytics on gym trends, occupancy patterns, and recommended actions.
- Detailed Reporting (UC-12): Allows authorized managers to generate and export custom reports on attendance, revenue, and facility utilization.
- System Administration (UC-13): A comprehensive console for administrators to manage all system aspects, including user roles, pricing plans, facility configurations, and hardware settings.

Business System Options Comparison

BSO 1: Essential Digitalization & Access Control

- Description: This option addresses the most critical flaw: the disconnect between online payments and gym access. It implements a mandatory QR-Based Access Control (UC-01-04) system. This forces digital attendance logging and, crucially, validates the online subscription (UC-02, UC-05) at the door before granting entry. It replaces the physical sign-in book and ensures only paying members enter.
- Solves Current Pain Points:
    - Eliminates the risk of members entering with expired/without subscriptions.
    - Automates and mandates attendance tracking, replacing the optional paper log.
    - Provides a definitive record of member visits.
- Pros:
    - Highest Impact for Cost: Directly stops revenue leakage.
    - Faster Member Entry: Scans are quicker than manual sign-in.
    - Foundation for Future Systems: Establishes the core digital member database.
- Cons:
    - Requires investment in hardware (QR scanners/tablets for entrance).
    - Members must adapt to using phones/QR codes instead of just walking in.

BSO 2: Operational Efficiency & Member Portal

- Description: This option builds on BSO 1 by digitizing the other major paper-based processes. It adds the Member Self-Service Portal featuring Trainer Appointments (UC-06) and Digital Workout Scheduling (UC-10). This eliminates

the outdated "generic workout card" and allows members to book and manage their routines via an app, reducing constant instructor admin tasks.

- Solves Current Pain Points:
    - Replaces physical workout cards with dynamic, updatable digital plans.
    - Streamlines the ad-hoc booking process, reducing scheduling conflicts and instructor workload.
    - Provides a clear value proposition to members (convenience), improving retention.
- Pros:
    - Significantly reduces administrative burden on instructors.
    - Improves member satisfaction and engagement through modern self-service tools.
    - Creates a new structured data source (appointment history).
- Cons:
    - Higher development cost than BSO 1.

BSO 3: Integrated Growth & Insights Platform (Recommended)

- Description: This is the complete system as proposed. It includes BSO 1 and 2 and adds the data analytics layer. This introduces Progress Tracking for members (UC-07, UC-08) and Insights & Reporting for management (UC-11, UC-12, UC-13). This turns operational data into strategic assets.
- Solves Current Pain Points:
    - Replaces the non-existent "Manual Vitals Input" with a seamless digital progress journey, a powerful member retention tool.
    - Empowers "Top Management" and "Admin" with real-time insights into attendance trends, revenue, and facility usage, moving them from passive overseers to active managers.
    - Provides a complete 360° view of both business health and member engagement.

- Pros:
    - Maximizes ROI: Uses data to drive membership retention, personalized marketing, and operational decisions (e.g., staff scheduling during peak hours).
    - Creates a Competitive Advantage: Offers features (progress analytics, insights) that differentiate the gym from competitors.
    - Future-Proofs the Business: The integrated platform can easily scale to add new services.
- Cons:
    - Highest initial investment and complexity.
    - Requires the most cultural shift in how both staff and members interact with the gym.

Conclusion and Recommendation

- BSO 1 is a pragmatic first step focused purely on securing revenue and core process digitization.
- BSO 2 is a significant upgrade aimed at boosting operational efficiency and member satisfaction.
- BSO 3 is the strategic choice for long-term growth and market leadership, transforming the gym from a traditional facility into a data-informed, modern fitness service.

Feature Category	**Specific Feature / Use Case**	**BSO 1: Essential Digitalization & Access Control**	**BSO 2: Operational Efficiency & Member Portal**	**BSO 3: Integrated Growth & Insights Platform**
Core Access & Security	UC-01: Scan QR Code (Entry/Exit)	X	X	X
	UC-02: Validate Subscription	X	X	X
	UC-03: Authenticate User	X	X	X
	UC-04: Record Timestamp (Auto Attendance)	X	X	X
	UC-09: Staff Enter/Leave Gym	X	X	X
Membership Management	UC-05: Buy/Renew Subscription Online	X	X	X
Scheduling & Appointments	UC-06: Make a Trainer Appointment		X	X
	UC-10: Change/Modify Workout Schedule		X	X
Member Engagement & Tracking	UC-07: Mark Workout Done		X	X

	UC-08: View/Track Progress (Dashboard)		X	X
Business Intelligence & Admin	UC-11: View Insights (Analytics)			X
	UC-12: Access Detailed Reports			X
	UC-13: Manage Entire System (Admin Console)			X

Requirements analysis – Functional

Functional requirements specify the core operations and features a system must deliver to meet its intended purpose. They define what the system is expected to do and outline the services it should provide to users.

These requirements typically address:

1. System Capabilities – The essential functions and tools the system must support.
2. User Interactions – The ways in which users engage with the system, including inputs, outputs, and interface behavior.
3. Data Management – How the system processes, stores, retrieves, and secures information.

ID	**Requirement Category**	**Requirement Description**	**Priority**
FR-01	Access Control	The system shall allow a member to scan a QR code using a mobile device for identification.	High
FR-02	Access Control	The system shall validate the member's subscription status in real-time upon a scan attempt.	High
FR-03	Access Control	The system shall authenticate the user's identity based on the scanned QR code data.	High

FR-04	Access Control	Upon successful authentication and validation, the system shall send a signal to unlock the entrance door.	High
FR-05	Access Control	The system shall record a timestamp with member ID, event type (IN/OUT), and location for every entry and exit.	High
FR-06	Access Control	The system shall prevent duplicate "IN" events for a member who is already logged inside.	Medium
FR-07	Access Control	The system shall provide instructive error messages for failed scans, invalid subscriptions, or authentication failures.	Medium
FR-08	Access Control	The system shall allow staff members to enter/leave the facility, logging their access.	Medium
FR-09	Membership Management	The system shall allow a member to purchase a new	High

		subscription plan via the web or mobile app.	
FR-10	Membership Management	The system shall integrate with a payment gateway to process subscription payments.	High
FR-11	Membership Management	Upon successful payment, the system shall automatically activate the member's subscription and update their record.	High
FR-12	Membership Management	The system shall send a confirmation notification (e.g., email) to the member after a successful purchase.	Medium
FR-13	Scheduling	The system shall allow members to view available time slots and instructors for appointments.	Medium
FR-14	Scheduling	The system shall allow a member to book an appointment with a selected instructor	Medium

		for an available time slot.	
FR-15	Scheduling	The system shall verify a member's subscription entitlements before allowing an appointment to be booked.	Medium
FR-16	Scheduling	The system shall send notifications to both the member and the instructor upon appointment creation.	Medium
FR-17	Scheduling	The system shall allow members and instructors to modify or reschedule existing workout appointments.	Medium
FR-18	Scheduling	The system shall check for scheduling conflicts (e.g., trainer availability) before confirming any modification.	Medium
FR-19	Progress Tracking	The system shall allow a member to mark a scheduled workout or routine as completed.	Medium

FR-20	Progress Tracking	The system shall record a completion timestamp when a workout is marked as done.	Medium
FR-21	Progress Tracking	The system shall recalculate and update the member's progress statistics upon workout completion.	Medium
FR-22	Progress Tracking	The system shall display a dashboard to the member with historical workout data, trends, and visualizations.	Medium
FR-23	Administration	The system shall provide an admin console for managing facilities, user roles, pricing plans, and system settings.	Low
FR-24	Reporting	The system shall allow managers to generate and export detailed reports (e.g., attendance, revenue) in PDF/CSV formats.	Low
FR-25	Security	The system shall validate user roles and permissions	High

		before allowing access to any functionality.	
FR-26	Analytics	The system shall provide an insights dashboard for managers showing trends, occupancy patterns, and recommended actions.	Low
FR-27	Data Management	The system shall store data locally and sync later if a network connection is unavailable (offline capability).	Medium
FR-28	Audit	The system shall log all user and administrative actions for audit purposes.	Low

Requirements analysis – Non-Functional

Non-functional requirements define the standards and attributes that shape how a system performs, rather than the specific tasks it carries out. These requirements play a critical role in ensuring the system operates smoothly, efficiently, and remains dependable over time.

They typically emphasize:

1. Performance – How fast, responsive, and efficient the system is under various conditions.

2. Reliability & Availability – The system's ability to function consistently and remain accessible when needed.

3. Scalability & Maintainability – How well the system adapts to growth and how easily it can be updated or improved.

ID	**Requirement Category**	**Requirement Description**	**Priority**
NFR-01	Security	The system shall use secure, signed, or one-time-use QR codes to prevent replay attacks.	High
NFR-02	Security	The system shall implement rate-limiting on authentication attempts to prevent brute-force attacks.	High
NFR-03	Security	All sensitive data (passwords, payment info) shall be encrypted in transit (TLS 1.2+)	High

		and at rest (AES-256).	
NFR-04	Performance	The system shall respond to a QR scan and unlock door signal in under 2 seconds to prevent entrance delays.	High
NFR-05	Performance	The reporting module shall generate standard reports for a 30-day period in under 10 seconds.	Medium
NFR-06	Reliability & Availability	The core access control system shall have 99.9% uptime during gym operating hours.	High
NFR-07	Reliability & Availability	The system shall gracefully handle failure of a secondary service (e.g., if email service is down, queue notifications).	Medium
NFR-08	Usability	The member mobile app shall be learnable for a new user to perform basic tasks (check-in, view schedule) within 5 minutes.	Medium

NFR-09	Compatibility	The member mobile app shall be compatible with Android (version YY and above).	High
NFR-10	Compatibility	The system shall integrate with standard payment gateways (e.g., Stripe, PayPal) using common APIs.	High
NFR-11	Scalability	The system architecture shall support a 20% increase in member count without a loss in performance.	Low
NFR-12	Maintainability	The system shall be modular, allowing individual components (e.g., billing, auth) to be updated independently.	Low
NFR-13	Data Integrity	The system shall validate event ordering to prevent illogical records (e.g., two consecutive "IN" events).	Medium

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